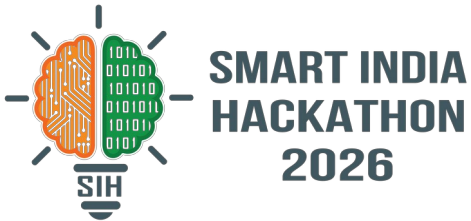


SMART INDIA HACKATHON 2026

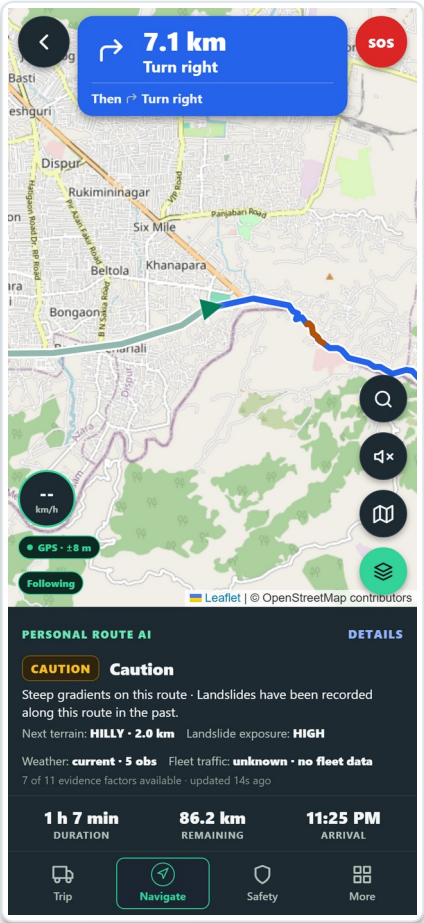
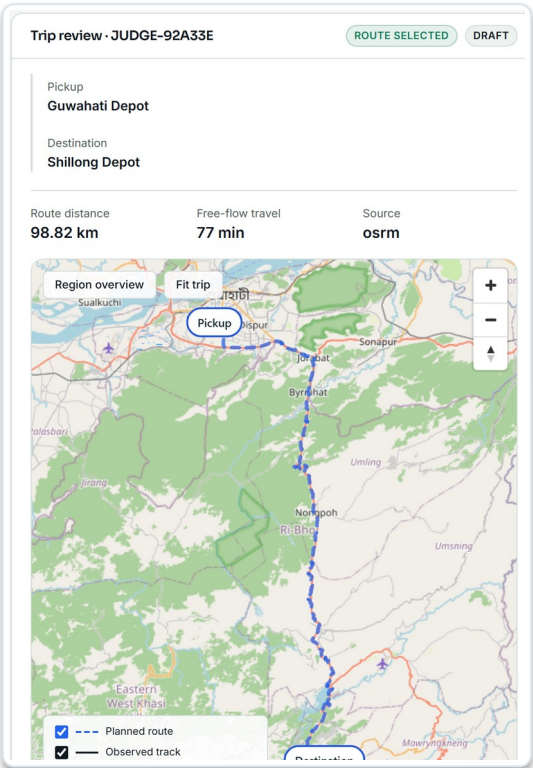


RASTA AI

Route Intelligence for Essential-Supply Logistics in North Eastern India

THE ASK	SIH26002		PROBLEM STATEMENT ID
	MDoNER · Smart Automation · Software		
THE RECORD	PROBLEM STATEMENT TITLE		AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)
	THEME		Smart Automation
	PS CATEGORY		Software
	TEAM ID		17 (internal group no.)
	TEAM NAME		NER-AI LOGISTICS
	SHORTEST IS NOT ALWAYS USABLE.		
THE POINT	“Can this essential-supply truck reliably use this corridor now?”		
	End-to-end working prototype · physical-phone validated · live demo corridor Guwahati → Shillong		

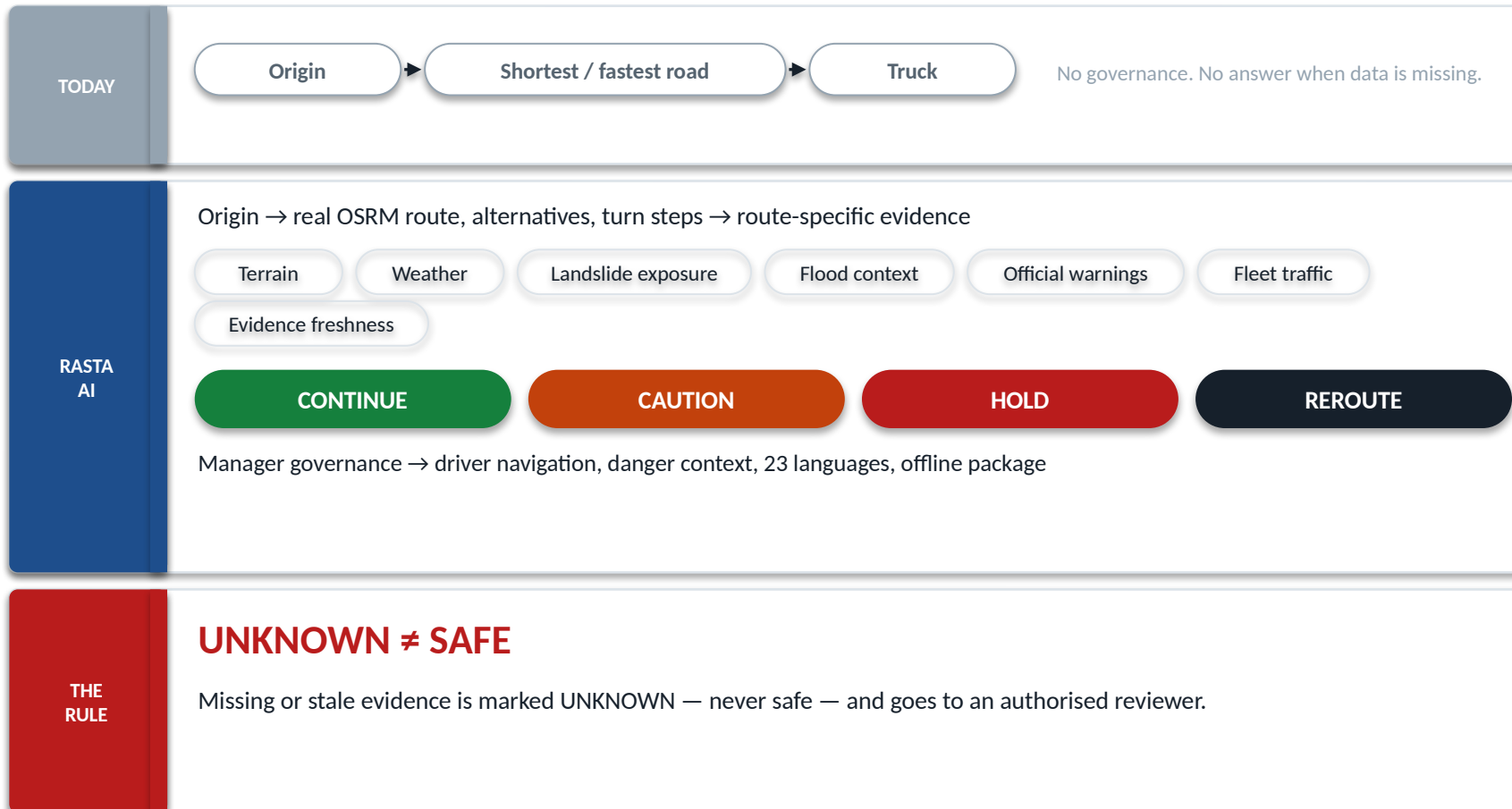
LIVE WORKING SYSTEM



Manager route review · Driver navigation — real screens

Not the shortest road. **The road that is operationally usable now.**

MANAGER • CHECK CONDITIONS

**REQUIRES REVIEW**

Hazard evidence is incomplete or elevated. An authorised reviewer must review this route before selection. Refresh conditions after review.

Landslide data is not available - no source is connected. · Steep gradients on this route · Landslides have been recorded along this route in the past. · The landslide inventory ends in 2017 - recent years are not covered. · River levels near this route are close to their recent average. · No official disaster alert names a district on this route right now. · No fleet truck has driven this road in the last fifteen minutes, so traffic is unknown.

Unavailable: Fuel range, Landslide, Road quality, Truck restrictions

Evidence coverage · Weather AVAILABLE · Terrain AVAILABLE · Warnings AVAILABLE · Traffic UNKNOWN

TERRAIN**0.5 KM STEEP**

49–1436 m · 1975 m climb · steepest 10.3%
flat 61.8 km · rolling 29.6 km · hilly 7.0 km · steep 0.5 km
Copernicus DEM GLO-90 via Open-Meteo + SRTM 30 m via OpenTopoData · 100% of 198 samples answered

LANDSLIDE HISTORY**HIGH EXPOSURE**

22 recorded landslides within 5 km of the road, nearest 0.4 km
6 more nearby placed too imprecisely to count · HIGH is three or more; MODERATE is one or two
NASA Global Landslide Catalog · inventory 2007–2017 — aged, recent years not covered

RIVER LEVELS**NORMAL**

Discharge at 5 river cells along the corridor, highest 1.0× its own 30-day mean · GloFAS via Open-Meteo, 2026-09-14. A level, not a flood forecast and not a road-closure claim.

OFFICIAL ALERTS**NONE ON CORRIDOR**

NDMA SACHET CAP feed, 99 alerts nationwide at 10:13:54 PM · placed by district name (Dispur, East Khasi Hills, Kamrup Metropolitan, Myllem, Ri-Bhoi, Umling, Umsning) · 4 more active elsewhere in the corridor states, not placeable on this road.

FLEET TRAFFIC**UNKNOWN**

Real screen · evidence · freshness · UNKNOWN

TECHNICAL APPROACH

**VERIFIED
DATA****WEATHER**

Open-Meteo · MET Norway

TERRAINCopernicus DEM ·
OpenTopoData**WARNINGS**

NDMA SACHET (CAP)

FLOOD

GloFAS

LANDSLIDE HISTORY

NASA GLC 2007–17

TRAFFIC

fleet GPS fixes

DECISION**ROUTE-SPECIFIC EVIDENCE · FastAPI backend**

every factor sampled along the selected road · PostgreSQL + PostGIS (Supabase) · OSRM routes, alternatives, turn steps

DETERMINISTIC SAFETY POLICY

11 factors · reason codes · UNKNOWN ≠ SAFE · reviewer authorisation when hazard data is missing

CONTINUE**CAUTION****HOLD****REROUTE**

MANAGER REVIEW

DRIVER NAVIGATION

ROUTE-AHEAD 60 s

CONDITIONS CHANGE?

REASSESS

**BUILT
WITH****CLIENTS** Manager console (React + TS) · Driver app (React Native / Expo, Android) · one login, server-decided role, offline package**STACK** FastAPI · PostgreSQL / PostGIS (Supabase) · React · React Native / Expo · OSRM · OpenStreetMap · Gemini**EVIDENCE** Open-Meteo / MET Norway · NDMA SACHET · GloFAS · NASA historical landslides · OpenTopoData / Copernicus DEM

Re-scored every 60 s on the road ahead. A reroute is proposed to the manager, never applied silently.

WHO DECIDES**DETERMINISTIC SAFETY ENGINE**

makes every operational decision

GEMINI / OPENROUTER (LLM)

explains the decision in the driver's language — never decides

EXPERIMENTAL ML

research channel only; held outside routing by the safety gate

GOVERNANCE CHAIN

Verified evidence



Deterministic safety policy



Manager governance



Driver action

Not a concept. A working end-to-end prototype.

LIFECYCLE

PLAN

REVIEW

DISPATCH

ACCEPT

TRUCK VERIFY

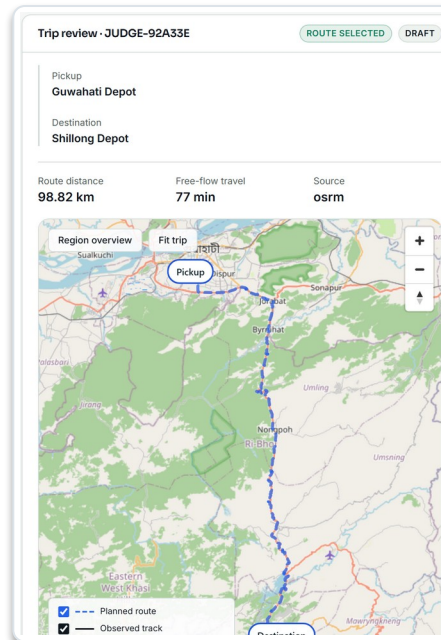
NAVIGATE

OFF-ROUTE

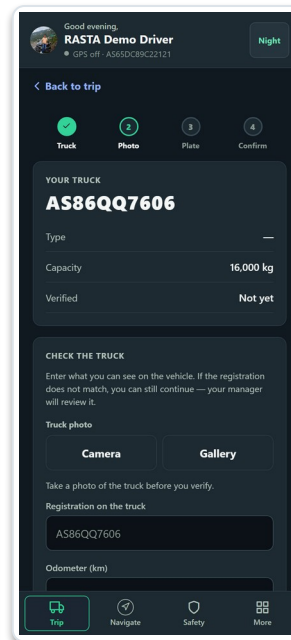
MANAGER REROUTE APPROVAL

DELIVERY

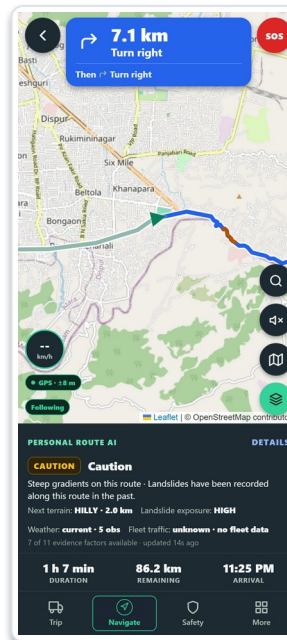
EVIDENCE



Manager · route review



Driver · truck check



Driver · navigation

Run end to end on a physical Android phone, hosted backend, real road.

12/12 physical judge-flow steps

9/9 physical role-flow steps

1138+ backend · 621+ driver · 170+ manager tests

PHYSICAL ANDROID · CERTIFIED

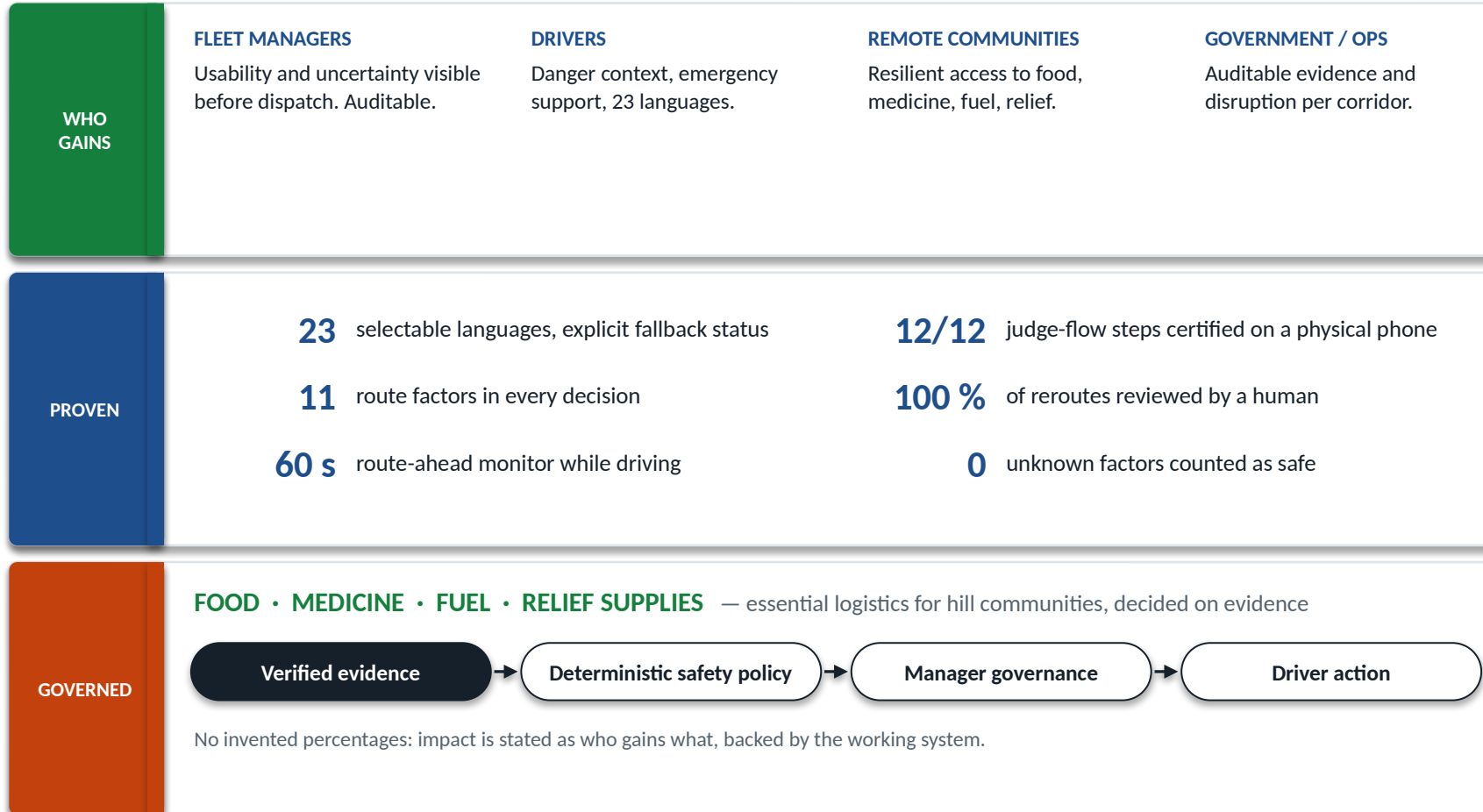
Canonical demo

Guwahati → Shillong · ≈ 98.8 km real route

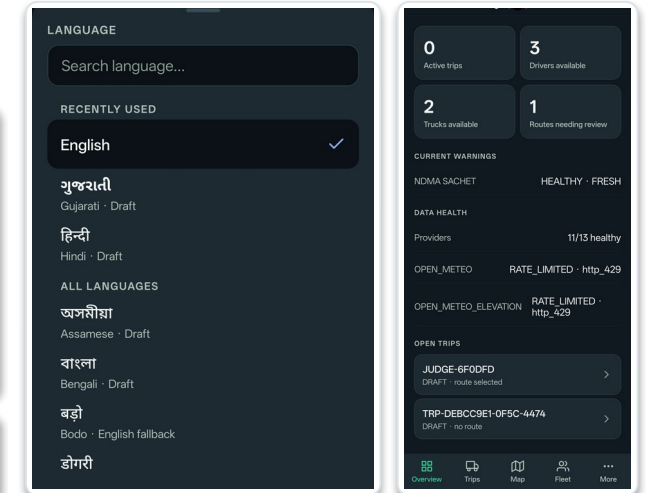
KNOWN LIMITS → MITIGATION Cold start → warm /health first · rate limits → cached evidence + explicit UNKNOWN · push needs Firebase → in-app alerts work · experimental ML → outside routing

IMPACT AND BENEFITS

RASTA AI does not ask only “Which road is shortest?” It asks “Can this truck reliably use this corridor now?”



REAL SCREENS • APK 1.0.18



Driver • 23 languages

Manager • mobile

APK 1.0.18. One login, server-decided role. Every language shows its status — Verified, Draft or English fallback.

RESEARCH AND REFERENCES

SOURCES

ROUTING / PLACES

OSRM • OpenStreetMap • Nominatim

Road geometry, alternatives, turn steps.

WEATHER / TERRAIN

Open-Meteo • MET Norway • Copernicus DEM • **OpenTopoData**

Forecast along the road; climb, gradient, steep segments.

OFFICIAL / FLOOD

NDMA SACHET (CAP) • Copernicus GloFAS

Official warnings with freshness; discharge vs 30-day mean.

LANDSLIDE HISTORY

NASA Global Landslide Catalog

Historical exposure, 2007–2017, ≤5 km accuracy.

AI ASSISTANCE

Google Gemini

Explains in the driver's language. Never safety authority.

PROBLEM STATEMENT

MDoNER • SIH26002

Smart logistics and accessibility intelligence for the NER.

EXPERIMENTAL LANDSLIDE RESEARCH

95 %

RECALL

NER geographic holdout • 154 events • no NER row in training

SAFETY GATE

NOT DEPLOYED

FPR 50 % too high for production deployment



VALIDATION BEFORE AUTOMATION. Recall was not enough. Held-out testing exposed the false-positive rate. The gate keeps it out of routing.

Precision 0.32 • F1 0.48 • PR-AUC 0.55 • ROC-AUC 0.83 • Brier 0.12 — calibrated logistic regression, NASA GLC + ERA5-Land rain + DEM slope